



Understanding the Role of Message Frames on African-American Willingness to Participate in a Hypothetical Diabetes Prevention Study

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The objective of this research was to evaluate the impact of message framing (e.g., highlighting health disparities vs. progress toward reducing disparities) on willingness to enroll in a hypothetical research study. African-American (AA, $n = 1513$) and White ($n = 362$) adults completed an online survey about diabetes, health behaviors including physical activity, and attitudes about research. AA participants were randomized to view a general message (same message as provided to all White participants) or 1 of 4 alternate messages that framed the need for people to participate in research in terms of race and/or health disparities. Among AAs, there were no differences in willingness to enroll in the study by message frame. However, individual characteristics including younger age, female sex, attitudes about research, a sense of obligation, and community responsibility were significant predictors of willingness to enroll in the study. AA participants who received the general message were equally willing as White participants to enroll in the study. Highlighting race and health disparities in study recruitment materials may not be needed to increase interest among AAs. Factors beyond race appear to be stronger motivators for participation. Unlike previous research, racial framing did not suppress motivation to enroll in our hypothetical study.

Historically, African-Americans have been underrepresented in health research studies across medical conditions and behaviors (Castillo-Mancilla et al., 2014; Ford et al., 2008; McIlvane, Baker, Mingo, & Haley, 2008; Siddiqui et al., 2015; Yancey, Ortega, & Kumanyika, 2006; Zhang, Tsang, Wijesundera, & Ko, 2013). Although there is growing evidence that African-Americans are willing to participate in health research when eligible and invited (Wendler et al., 2006), barriers to African-American participation in research include lack of knowledge, co-morbidities, distrust of the medical system, and disparities in the offer of and communications about research opportunities (R. F. Brown et al., 2013; Ford et al., 2008; Langford et al., 2014; Otado et al., 2015; Schmotzer, 2012). The Theory of Planned Behavior is one behavior theory that has been applied to the context of health research participation (Lucas, Hayman, Blessman, Asabigi, & Novak, 2015; Quinn et al., 2010; Yang et al., 2009). The theory assumes that an individual's attitudes, subjective norms, and perceived behavioral control create behavioral intention, which in turn is linked to performing a behavior such as joining a health research study (Ajzen, 1991). Numerous

behavioral theories including Theory of Planned Behavior, Health Belief Model, and Social Cognitive Theory all postulate, though with varying terminology, that an individual's perceived benefits, barriers, and normative beliefs should predict one's willingness to participate in health research (Glanz, Rimer, & Viswanath, 2015).

Moreover, despite growing attention to the importance of effective communication in patient care and health promotion campaigns for the general public (National Cancer Institute, 2004; Spooner, Salemi, Salihu, & Zoorob, 2016), the role of health communication in promoting health research studies has not been adequately explored (Jenerette & Mayer, 2016; Lederer et al., 2015; Spooner et al., 2016). Specifically, more information is needed to understand how, if it all, the framing of race and health disparities information may impact African-Americans' willingness to participate in health research studies. To date, the literature on message framing has been largely been grounded in Prospect Theory which posits that how people choose between options with varying levels of risk is influenced by how these options are framed (Kahneman & Tversky, 1979). For example, "gain-framed" messages highlight the positive benefits of engaging in or avoiding certain behaviors, while "loss-framed" messages highlight the negative consequences or risks associated with certain behaviors (Akl et al., 2011; Rothman & Salovey, 1997). Some studies have shown that

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gain-framed messages are more likely than loss-framed messages to encourage prevention behaviors (Gallagher & Updegraff, 2012; Latimer et al., 2008). Other studies have shown that African-Americans are more likely to respond positively to gain-framed messages compared with Whites (Lucas et al., 2015; Schneider et al., 2001), while other studies conducted solely with African-Americans have shown that participants respond more favorably to gain-framed messages than loss-framed messages (Nicholson et al., 2008; Satia, Barlow, Armstrong-Brown, & Watters, 2010). Regardless of the health communication strategy used, the idea of “contextualization” or framing a message in a way that is meaningful and personally relevant to the reader is critical to increase motivation to engage in a behavior (Hawkins, Kreuter, Resnicow, Fishbein, & Dijkstra, 2008).

Highlighting racial disparities or opportunities to reduce them is one strategy that has sometimes been used to appeal to potential study participants (S. D. Brown et al., 2012; Langford, Resnicow, & Beasley, 2015) and to motivate positive behavior change more broadly (Resnicow et al., 2004, 2005, 2000). This approach generally assumes that minorities will be more motivated to adopt a health behavior change or engage in a health research study if messages are written in a culturally relevant manner and/or include ethnically-focused statements (S. D. Brown et al., 2012). For example, messages that highlight disparities between particular racial/ethnic groups for a specific health condition may be assumed to be more motivating than general audience messages that acknowledge unique racial/ethnic differences. However, these assumptions were questioned by a study which showed that messages emphasizing health disparities in colorectal cancer (CRC) led to lower intentions of African-American adults to obtain CRC screening (Nicholson et al., 2008). In that study, Nicholson and colleagues evaluated the impact of four health messages on intentions to get screened. Their message frames included *impact* (CRC is an important problem for African-Americans), *disparity* (Blacks are doing worse than Whites), *disparity over time* (Blacks are improving, but less than Whites), and *progress* (Blacks are improving over time). They found that messages that highlighted health disparities could be detrimental. That is, participants exposed to disparity messages that compared Blacks and Whites had more negative emotional reactions to the information and were less likely to want to be screened for CRC. In contrast, messages that highlighted progress in Blacks over time significantly increased intentions to get screened. Furthermore, in a sample of South Africans, Evangeli and colleagues examined the role of message framing on willingness to participate in a hypothetical HIV vaccine trial and found that gain-framed messages were more likely than loss-framed messages to be associated with willingness to participate in the trial (Evangeli, Kafaar, Kagee, Swartz, & Bullemor-Day, 2013). Despite these intriguing findings, it is unclear how disparity focused messages and framing effects may generalize to health behaviors beyond CRC and research opportunities beyond HIV.

The primary objective of our study was to explore how, if at all, framing health disparities information might impact African-Americans’ willingness to participate in a hypothetical diabetes prevention study. Five message frames were tested in our study: impact, disparity no action, disparity with action, progress, and

general. Diabetes was selected as the topic for the study because it is a common health condition for which a high proportion of survey samples could relate to, has large racial disparities, and is a health condition that can be mitigated with lifestyle change including physical activity (American Diabetes Association, 2017). Based on prior studies and behavioral theory principles, we hypothesized that among African-Americans: 1) willingness to enroll in the hypothetical study would be greatest for the *progress* message, 2) willingness to enroll in the study would be lowest in the *disparity no action* message, and 3) the *disparity with action* message might mitigate the potential negative effects of presenting disparity messages without an action step. For African-American and White participants who received the general message without reference to race or health disparities, we hypothesized that there would be no difference in willingness to enroll in the study. Finally, we hypothesized that willingness to enroll in the hypothetical study would be moderated by individual characteristics including age, sex, education, and attitudes about research.

Methods

We recruited participants through Survey Sampling International (SSI). SSI maintains a global panel of more than 11.5 million qualified respondents from certified sources in over 100 countries and approximately 5 million of these respondents are located in the United States (Survey Sampling International, 2015). SSI recruits members using multiple opt-in techniques including Internet banner ads, online recruitment methods, and random digit dialing. All members of SSI’s research panels complete a screening questionnaire to elicit detailed demographic information. SSI uses these data to select stratified random samples that match the US Census in terms of age, gender, income, and geographical location. Recruitment was managed through SSI’s sample management platform, Dynamix®. Participants registered with SSI were sent a daily email inviting them to the SSI survey platform. Once there, the Dynamix® system used a process of refinement to find people who were active in the system, looking to take a survey, and who meet the sampling criteria. Direct email invites were used as a last resort, only if sampling quotas were not being met.

Participants were eligible for the study if they 1) self-identified as African-American/Black or White, 2) were 30 years of age or older, and 3) resided in the United States. We chose 30 as the lower age cut off because diabetes is more likely to be diagnosed as people age (in 2011, <5% of incident diabetes cases were diagnosed between 18 and 29), and we wanted the hypothetical scenario to be a realistic concern for respondents (Centers for Disease Control and Prevention, 2015). Given that our goal was to evaluate potential differences in willingness to participate in the hypothetical study in a large sample of African-Americans, our target sample was 1500 African-Americans and 300 Whites for comparison with the general message frame. Within the target numbers, we wanted the sample to be evenly divided in terms of age (e.g., <59 and 60+) and gender. The study was programmed with Qualtrics® survey platform and administered through SSI’s online panel. Survey data collection began on 17 October 2014 and ended on 28 October 2014 once target numbers were reached.

With regard to the SSI’s randomization procedure, confidence limits based on the sample size within the sampling frame described above are as follows: if the study was repeated using the same set of sources, 19 times out of 20 we would expect to get a result within $\pm 3\%$. SSI’s blend algorithm is utilized which monitors the sample for consistency on a number of personality and psychographic measures drawn from academic literature. Once within SSI’s system, participants are matched with an available survey using multiple points of randomization. Security checks and quality verifications are used on all sources before the participant can begin any survey (personal communication, F. Markowitz, February 23, 2017). The survey process for all participants was completely anonymous. At the conclusion of each survey, participants were routed back to SSI, which provided the incentives. Approximately 2,232 adults from SSI started the survey and 1,894 participants completed the survey (Figure 1). Nineteen participants did not meet the inclusion criteria, leaving a final sample of 1,875.

Study Design

African-American participants were randomized to view either a general message without reference to race or health disparities or 1 of 4 alternate messages that framed the issue in terms of race and health disparities. White participants were assigned to view the same general message as African-Americans in the control condition and were not randomized to any other message type. African-American and White participants first read a message frame at the beginning of an online web page about the hypothetical study. Next, participants read specific details about the study including the eligibility criteria, required tasks, and compensation. Lastly, participants answered questions about their willingness to enroll in the hypothetical study, as well as questions about their

general health status, physical activity behaviors, and attitudes about health research. The average time for participants to complete the survey session was approximately 20 minutes. The university’s Institutional Review Board approved the study.

Message Frames

The five message frames evaluated in this study included impact, disparity no action, disparity with action, progress, and general. For the disparity with action message, we defined “action step” by including a sentence at the end that read, “By participating in an exercise research study, you can help us better understand how to reduce the rate of chronic diseases among African-Americans.” The communication goal of the action step was to prompt readers to think about how participating in research could potentially contribute to improving African-American health as a way to provide contextualization. The disparity with action step, progress, and general messages was conceptually similar to a gain frame message, while the impact and disparity no action messages were conceptually similar to that of a loss frame message. The messages were developed by our study team, which included experts in health communication, risk communication, and behavior change. The full messages can be seen in Table 1.

Hypothetical Study Scenario

The hypothetical study was entitled, *Preventing and Managing Type 2 Diabetes with Physical Activity*. Participants were given the following prompt at the beginning of the scenario, “Imagine that your doctor told you that, because your A1c blood level is higher than 5.7%, you could take part in the following study. An A1c of higher than 5.7% means that you might develop diabetes in

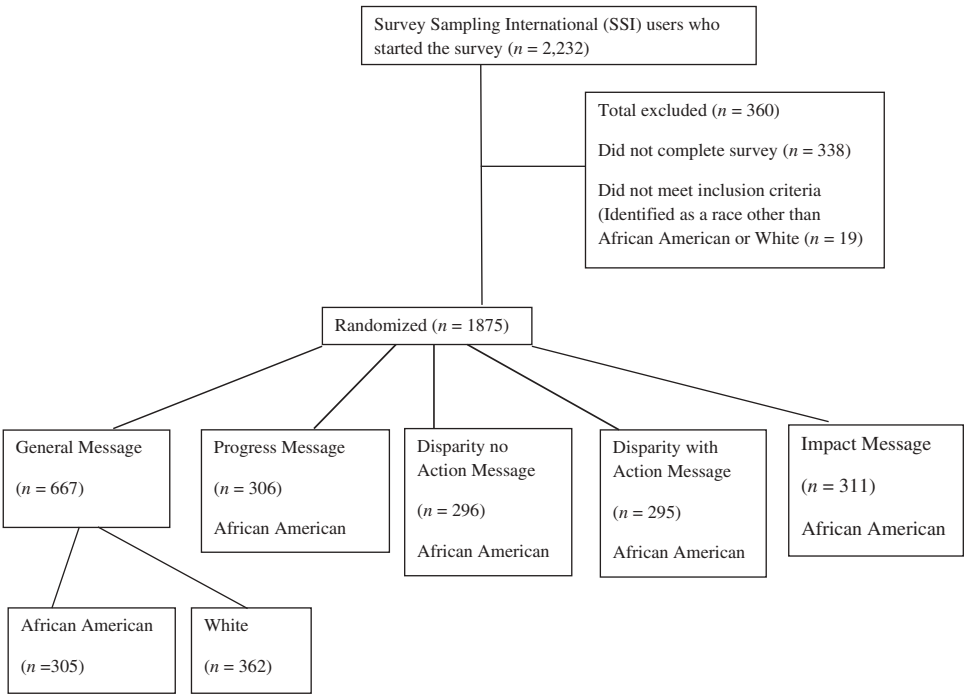


Fig. 1. Study flow diagram.

Table 1. Message frames.

Message frame and content
Impact Every year, thousands of African-Americans die from chronic diseases like diabetes. Lack of exercise is a major reason for these high numbers. More research is needed to understand how to get more African-Americans to exercise.
Disparity no action African-Americans die from chronic diseases like diabetes at a much higher rate than Whites. This is in part because African-Americans exercise less than Whites. African-Americans are also underrepresented in physical activity research. More research is needed to understand how to get more African-Americans to exercise.
Disparity with action step African-Americans die from chronic diseases like diabetes at a much higher rate than Whites. This is in part because African-Americans exercise less than Whites. African-Americans are also underrepresented in physical activity research. More research is needed to understand how to get more African-Americans to exercise. By participating in an exercise research study, you can help us better understand how to reduce the rate of chronic diseases among African-Americans.
Progress African-Americans are increasing their exercise levels. This should lead to lower rates of chronic diseases like diabetes. African-Americans are also increasing their participation in research studies. In fact, recent studies have shown that African-Americans are quite willing to participate in research and do so when invited. More research is needed to understand how to get even more African-Americans to exercise.
General Americans are increasing their exercise levels. This should lead to lower rates of chronic diseases like diabetes. Americans are also increasing their participation in research studies. In fact, recent studies have shown that Americans are quite willing to participate in research and do so when invited. More research is needed to understand how to get even more Americans to exercise.

the very near future. The next screen will have details about the study.” In the scenario, adults 18 and over who were at risk for getting diabetes or had diabetes were eligible to join the study. Participants were told that they would be randomized to an in-person physical activity group lead by a health coach or an online physical activity group lead by a health coach, and that they would be compensated up to \$75 for their participation. The full scenario can be found in the [appendix](#).

Measures

Primary Outcome: Willingness to Enroll in the Hypothetical Diabetes Prevention Study
The primary outcome was worded as, “Given what you know about this study, how likely are you to enroll?” Response options ranged from not at all likely (1) to very likely (5).

Secondary Outcomes: Willingness to Participate in Other Types of Health Research Studies

Given that willingness to participate in research may differ by the types of research and tasks involved, we asked a broader set of questions that reflected other research scenarios in which people may be invited to participate. These items were worded as, “There are many types of health studies. How likely would you be to participate in a study if... 1) it only asked questions about your health, 2) researchers wanted to see your medical records, 3) you had to give a blood sample, 4) you were asked to give a sample for genetic studies (swab of mouth), 5) you might have to take medicine, 6) you were asked to stay overnight in a hospital or clinic, 7) you might have to use medical equipment, and 8) you didn’t get paid.” Response options ranged from not at all likely (1) to very likely (5).

Predictor Variables

Demographic variables including age, education, and gender, as well as the five message frames were included in all multivariate linear regression models. The items, *All people should feel obligated to participate in a research study* (obligation), and *It is my responsibility to my community to participate in this type of research* (community responsibility) were also included in the linear regression models and developed by the study team. Finally, we created a composite score for attitudes about research using survey items from the Research Attitudes Questionnaire or RAQ (Rubright, Cary, Karlawish, & Kim, 2011). In short, the RAQ is an 11-item scale that measures how favorably or unfavorably individuals view biomedical research (alpha = 0.81). Higher scores indicated more positive attitudes about research.

Data Analysis

First, a one-way ANOVA was conducted to compare the effect of message frame on African-Americans’ willingness to enroll in the hypothetical study. Next, we used multivariate linear regression to determine whether the five message frames were associated with willingness to enroll in the hypothetical study among African-Americans, with the general message used as the reference category and adjusting for age, education, gender, attitudes about research, obligation to participate in research, and community responsibility to participate in research. Pairwise comparisons of all messages were evaluated using the *pwcompare* command in Stata; however, none of the pairings was statistically significant. Finally, using an independent samples *t*-test, we evaluated willingness to participate in the hypothetical study among African-Americans and White participants who received the general message without reference to race or health disparities. Independent samples *t*-tests were also used to evaluate differences in African-American and White willingness to participate in other types of health research studies (e.g., studies about genetics or those involving medication). All analyses were run in STATA version 14.0.

Results

As shown in [Table 2](#) and consistent with our target quota, participants were African-American (81%) and White (19%)

Table 2. Sample characteristics ($N = 1875$).

Characteristic	Mean	%	<i>n</i>
Attitudes about research score ^a	3.4		
Message frames			
General (ref)		36	667
Progress		16	306
Disparity no action		16	296
Disparity with action		16	295
Impact		16	311
Education			
High school or less (ref)		18	338
Postsecondary/trade		45	839
4 year college +		37	693
Race			
White (ref)		19	362
Black		81	1513
Age (30–85)	54		
30–59 (ref)		52	960
60+		48	878
Gender			
Males (ref)		48.85	913
Women		50.94	952
Transgender		0.21	4

^aScale of 1–5, *strongly disagree* to *strongly agree*; 13 items, $\alpha = .8$.

adults from across the United States with a mean age of 54. Approximately half of the sample was between the ages of 30–59 (52%), with the other half being over 60 (48%). With regard to gender, men comprised 48.8% of the sample, women comprised 50.9%, and transgender individuals comprised 0.21%. Most participants had post high school education, with only 18% having a high school degree or less, 45% having postsecondary or trade school training, and 37% having at least a bachelor's degree. Among African-Americans, there was no significant difference in education, gender, or age by message frame. When comparing African-Americans to Whites by education, gender, and age, there was a significant difference in educational attainment, with Whites having higher levels of education than African-Americans. We did not collect information about income in this study.

The effect of message frame on African-Americans' willingness to enroll in the study was not significant, $F(4,1502) = 0.56$, $p = .68$. Mean willingness by message frame was as follows: progress ($M = 3.48$), disparity no action ($M = 3.6$), disparity with action ($M = 3.46$), general ($M = 3.44$), and impact ($M = 3.46$). As shown in Table 3, the multivariate linear regression model controlling for attitudes about research, age, gender, obligation to participate in research, and community responsibility to participate in research yielded no differences among African-Americans for willingness to enroll in the hypothetical study by message frame (i.e., impact, disparity no action, disparity with action, progress, and general). An independent samples *t*-test was conducted to compare mean willingness to enroll in the hypothetical study between African-Americans and Whites. As shown in Figure 2, there was not a significant difference in willingness to enroll for African-Americans

($M = 3.44$, $SD = 1.47$) and Whites ($M = 3.32$, $SD = 1.45$); $t(663) = -1.07$, $p = 0.28$.

The multivariate linear regression models demonstrated that individual characteristics were associated with willingness to enroll in the hypothetical study (regardless of message type). For example, individual attitudes about research significantly predicted African-American willingness to enroll in the study ($B = 0.38$, $p < .001$). That is, as positive attitudes about research increased, willingness to enroll in the study also increased. Compared to African-Americans aged 30–59, African-Americans over the age of 60 were significantly less willing to enroll in the hypothetical study ($B = -.37$, $p < .001$). With regard to age, some trends were similar between younger and older African-Americans. Specifically, attitudes about research were significantly associated with willingness to enroll in the hypothetical study for African-Americans aged 30–59 ($B = 0.32$, $p = .001$) and African-Americans aged 60+ ($B = 0.46$, $p < .001$).

Moreover, African-Americans who expressed a greater sense of obligation to participate in a research study ($B = 0.16$, $p < .001$) and greater community responsibility for participating in research ($B = 0.36$, $p < .001$) were significantly more willing to enroll in the hypothetical study than those with lower scores. Similar trends were also observed for White participants who received the general message frame (Table 4). That is, White participants aged 60+ were less willing to enroll in the study compared to White participants aged 30–59. Additionally, attitudes about research and community responsibility were associated with willingness to enroll in the hypothetical study among Whites. However, the item related to feeling obligated to participate in a research study was not significant for Whites.

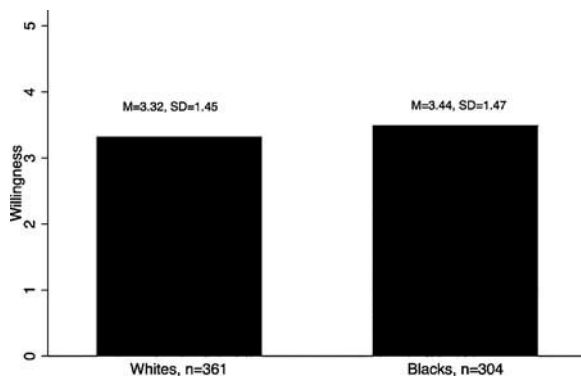
Discussion

In this study, we evaluated five message frames on African-Americans' willingness to participate in a hypothetical diabetes prevention study, controlling for age, gender, education, and attitudes about research. Our message frames included a general message without reference to race or health disparities (same message provided to White participants) or 1 of 4 alternate messages that framed the issue in terms of race and health disparities.

In contrast to our hypotheses, we did not find an effect of message framing among African-Americans in our study. This finding is different from previous research by Nicholson and colleagues who found that emphasizing health disparities in the context of colorectal cancer among African-Americans decreased intentions to get screened for CRC and elicited negative emotional reactions to the health information presented. The difference in our findings (i.e., no effect of message frame) compared to Nicholson and colleagues' findings (significant effect of message frame) may be explained in a variety of ways. First, it is possible that our hypothetical research study was not perceived as threatening or concerning enough to elicit different reactions by message frame. That is, our invitation to participate in a hypothetical diabetes prevention study may have been more palatable than engaging in a personal health behavior like colorectal cancer screening where *both* the screening behavior (e.g., colonoscopy) and potential outcome (cancer diagnosis) could

Table 3. African-Americans: Predictors of willingness to enroll in hypothetical study ($N = 1419$).

	β	p value	95% Conf. lower	95% Conf. upper
Attitudes about research composite score	.38 (.06)	.00	0.25	.51
Message frames				
General (ref)				
Progress	.01 (.1)	.87	-.18	.21
Disparity no action	.17 (.1)	.09	-.02	.37
Disparity with action	-.02 (.1)	.78	-.23	.17
Impact	.06 (.1)	.5	-.13	.26
Education				
High school (ref)				
Postsecondary/trade	.03 (.09)	.72	-.14	.21
4 year college +	.01 (.09)	.86	-.17	.2
Age				
30–59 (ref)				
60+	-.37 (.06)	.00	-.5	-.24
Gender				
Males (ref)				
Females	.13 (.06)	.03	-.006	.26
Transgender	-.85 (.61)	.16	-2.06	.35
All people should feel obligated to participate in a research study	.16 (.03)	.00	.1	.22
It is my responsibility to my community to participate in this type of research.	.36 (.03)	.00	.3	.43

**Fig. 2.** Willingness to enroll in hypothetical study by general message frame ($N = 665$): $t(663) = -1.07$, $p = .28$.

raise immediate health concerns. As observed by the responses to willingness to participate in other types of research studies (e.g., medication studies, genetic studies), we may have observed different responses if the hypothetical scenario focused on an invasive procedure like surgery or medication. Second, it is possible that racial differences and preferences for participation in health research or related health services have changed over time or that racial differences are only elucidated in certain situations. Another study examining message framing on African-American women's intention to participate in health-related research also found no effect of message framing on enrollment into ResearchMatch, a national volunteer registry

for health research studies (Balls-Berry et al., 2016). Although not specific to race or health disparities, researchers in South Africa did find an effect of message framing in the context of a hypothetical HIV vaccine study; greater willingness to participate in their hypothetical study was associated with gain-framed messages compared to loss-framed messages (Evangeli et al., 2013).

Given the scant literature on the benefits and risks of using race and health disparity information to promote health behavior change among African-Americans, our study is important because it indicates neither benefit nor harm from presenting race and disparity information to African-Americans in the context of a hypothetical diabetes prevention study. Moreover, among African-Americans and Whites who received the general message frame without reference to race, there was no significant difference with regard to willingness to participate in the hypothetical study.

It should be noted that the efficacy of using health message frames to increase participation in health research studies is still under investigation, with few published studies about this topic (Balls-Berry et al., 2016; S. D. Brown et al., 2012; Evangeli et al., 2013). More research is needed to explore how, if at all, message framing should be used in the context of clinical trial promotion and recruitment across various types of health research. Prospect Theory suggests that the way messages are framed impacts how people respond to information. While the messages frames in our study were not written to be gain-framed or loss-framed per se, one can argue that our impact and the disparity-no-action frames were conceptually similar to a loss frame, while the progress, disparity-with-action, and general

Table 4. White Americans. Predictors of willingness to enroll in hypothetical study. $n = 339$.

	β	p value	95% Conf. lower	95% Conf. upper
Attitudes about research composite score	.71 (.12)	.00	.46	.97
Education				
High school (ref)				
Postsecondary/trade	-.1 (.19)	.6	-.48	.27
4 year college +	.35 (.18)	.05	-.01	.71
Age				
30–59 (ref)				
60+	-.34 (.13)	.01	-.61	-.07
Gender				
Males (ref)				
Women	.15 (.13)	.23	-.1	.42
All people should feel obligated to participate in a research study	.09 (.07)	.21	-.05	.23
It is my responsibility to my community to participate in this type of research.	.3 (.07)	.00	-.15	.46

messages are conceptually similar to a gain frame or positive-framed message. Moving forward, highlighting race and health disparities for *group-level* study recruitment materials (i.e., targeting or audience segmentation) may not be necessary to increase interest in health research for African-Americans. Alternatively, developing recruitment messages based on *individual-level* attributes and preferences for information may hold promise (i.e., tailoring). For example, some research has shown that if ethnic identity (the degree to which a person identifies with a specific ethnic group) or racial pride is high, race-based messages may be effective for increasing receptivity to health information and/or promoting behavior change (Kreuter et al., 2003; Resnicow et al., 2009).

Regarding other individual-level characteristics evaluated in our study, age, attitudes about research, and community responsibility were significant predictors of willingness to enroll in the hypothetical study for both African-Americans and Whites. One interesting racial difference was the degree to which participants agreed with the statement, “All people should feel obligated to participate in a research study.” This item was significantly associated with willingness to enroll in our hypothetical study for African-Americans, but not for Whites. Although we had a smaller sample size for Whites, this finding may suggest that African-Americans may be more influenced by communal responsibility or have a stronger sense of collectivism compared to other racial groups (Coon & Kemmelmeier, 2001; Daly, Jennings, Beckett, & Leashore, 1995; Gaines et al., 1997), which may in turn influence their views about participating in health research. It is also possible that over time, African-Americans have increasingly become more open to participating in health research studies due to being disproportionately affected by chronic conditions like heart disease, cancer, and diabetes. Future messages emphasizing communal responsibility may hold promise for increasing uptake among African-Americans.

With regard to willingness to participate in other types of research studies, we did not find any differences between

African-Americans and Whites for studies that: 1) only asked questions about health, 2) where they had to give a blood sample, 3) involved genetics, 4) involved an overnight stay at a hospital or clinic, or 5) involved medical equipment. However, African-Americans were less willing than Whites to participate studies that required taking medication, where researchers wanted to see their medical records, and studies that involved no financial compensation. Related to expectations around compensation by race, a prior study found that African-Americans had the highest expectations regarding payment for a hypothetical study that was 1.5 hours long and required a blood draw (mean = \$81.60), followed by Hispanics (mean = \$75.15), and Whites (mean = \$61.68) (Cottler et al., 2013). This finding by Cottler would suggest that compared to other racial/ethnic groups, African-Americans would be less interested in participating in studies for which they were not paid at all or for studies in which they felt that compensation was not adequate. Furthermore, related to the use of medical records for research, Brown and Moyer also found that Blacks felt significantly less positive about the use of medical information for research compared with Whites (M. Brown & Moyer, 2010).

Limitations

We selected participants from a pool of people who registered to participate in Survey Sampling International's (SSI) online surveys and polls. Because we based the sample on those who initially self-selected for participation rather than a probability sample, no estimates of sampling error can be calculated. All surveys and polls administered through SSI may be subject to multiple sources of error, including but not limited to sampling error, coverage error, and measurement error. The data were not weighted to reflect the demographic composition of the general US population. Additionally, our SSI sample is not representative of the general population in several ways: 1) we manipulated the racial, age, and gender distribution to create a target

sample, 2) SSI consists of people who have access to the Internet and are comfortable taking online surveys, and 3) SSI consists of people who generally like taking surveys.

While the latter criteria may (in principle) be related to interest in health research study participation broadly, we cannot make this assumption with confidence. Our finding that African-Americans and Whites did not respond differently to messages that invite them to participate in research is not completely surprising. On one hand, we conducted our study in a sample of people who are generally open to the idea of research as evidenced by their participation in the SSI panel. On the other hand, we observed differences between African-Americans and Whites in their willingness to participate in other types of research studies; therefore, we cannot assume that every person in SSI's panel will automatically be willing to participate in every health research study presented to them in the future. Indeed, it is possible that we would have found different results if our study was implemented in a "real-world" setting of African-Americans and Whites from the general public.

Our hypothetical study mentioned that participants would be randomized to one of two groups, and the concept of randomization can be confusing for some people (Kodish et al., 2004; Krieger, Parrott, & Nussbaum, 2011). Although experts in health communication and behavior change developed the materials in the study, we did not pilot test our messages or the hypothetical scenario with participants; thus, we may have missed potential signs of confusion among readers. Although we attempted to have a similar structure and consistent statement in each message frame, the messages were not exactly the same in length or word count. Finally, we had a small sample size of Whites, which may explain why some trends were not significant for this group. Whites also had higher levels of education than African-Americans, which may have explained some of the racial differences in secondary outcomes.

Future Directions

Given the exploratory nature of health research studies and the complex motivations of individuals wanting to participate in such studies, our findings raise several questions for future investigation. First, further evaluation of which behavioral theory is best suited to explain health research participation might inform future efforts to promote various studies. Second, exploration of which health communication strategies best predict inquiries about and actual enrollment into health research studies is also needed. Third, the specific aspects of research that are of most concern to participants should be further evaluated (e.g., randomization, time commitment, intervention type). These concerns could vary by age, race, gender, and health condition. Finally, it is not clear if using a gain-frame vs. loss-frame approach in the context of health research participation is the most appropriate strategy as this method uses persuasion, whereas a more objective approach, like using decision aids, may enhance informed decision-making (Nathan, Marshall, Cooper, & Huang, 2016; Stacey et al., 2017). That is, with research, the actual benefits of an intervention are unknown until the study is completed; therefore, the idea that something will be gained or lost is

speculative. Moreover, if individuals choose not to enroll in a health research study, they will still have access to the current standard of care and will not necessarily be harmed by a lack of participation.

Conclusions

Highlighting race and health disparities in health research study recruitment materials may not be necessary to increase interest among African-Americans. Factors beyond race may be stronger motivators for participation. More research is needed to understand racial differences in willingness to participate in different types of health research studies. Finally, more research on when to use tailored messages that incorporate individual-level attributes versus targeted approaches that focus on group-level attributes may be useful for enhancing health research participation across racial/ethnic groups in the future (Kreuter & Wray, 2003; Schmid, Rivers, Latimer, & Salovey, 2008).

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Appendix: Hypothetical Scenario

Imagine that your doctor told you that, because your A1c blood level is higher than 5.7%, you could take part in the following study. An A1c of higher than 5.7% means that you might develop diabetes in the very near future.

Title of Study: Preventing and Managing Type 2 Diabetes with Physical Activity

The purpose of this study is to understand what motivates people to exercise and to assess differences in activity levels between two groups.

Who can participate?

Adults 18 and over, who are at risk for getting diabetes or have diabetes. “At risk” for getting diabetes means that your A1c is between 5.7% and 6.4%. Having diabetes means that your A1c is 6.5% or above. The A1c is a blood test that measures your average blood glucose (blood sugar) control for the past 2 to 3 months.

What's involved?

Study participants will be randomized into 1 of 2 groups:

an **in-person** physical activity group led by a health coach

OR

an **online** physical activity group led by a health coach

All participants will be asked to:

Complete an informed consent form. This form tells you about the risks and benefits of being in a research study. You must sign it before participating in a study.

Have your doctor say that you are in good enough health to be in the study (medical clearance).

Take 3 surveys about your current health situation, thoughts about physical activity, and opinions about clinical trials.

Attend 12 group physical activity sessions (in-person or online). Each session will last 1 hour.

Compensation

You will receive \$25 for each survey you complete (up to \$75).

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